Do Now · retrieval

Timed Paper 1 data and logic practice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which sort builds an ordered prefix by inserting a key?

Q2

How do you compensate for shifting a positive fractional mantissa left?

Q3

Simplify ¬(A ∨ B).

Q4

Find the decimal value of $0.1100_2 \times 2^3$.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Show positive first — Use exactly eight bits for +11, rather than padding after converting.

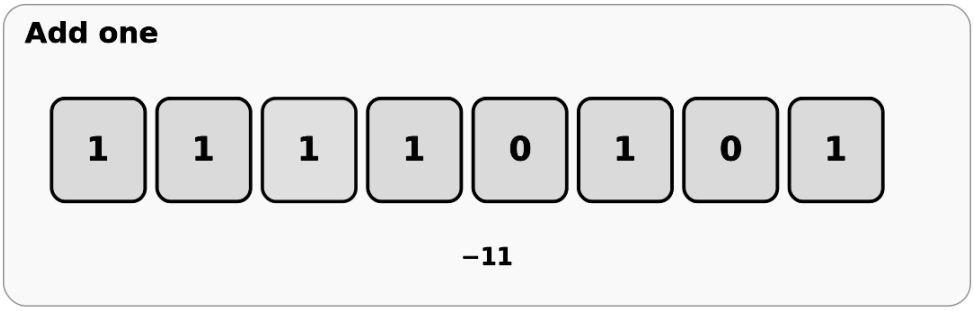
Core idea 2

Invert each bit — 00001011 becomes 11110100.

Core idea 3

Add one — 11110100 + 1 = 11110101 for −11.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

sign extension — Copy the sign bit into a new high-order position.

De Morgan — Swap AND/OR while negating components.

normalisation — Remove redundant leading mantissa sign bits.

mantissa — The fractional significant part.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

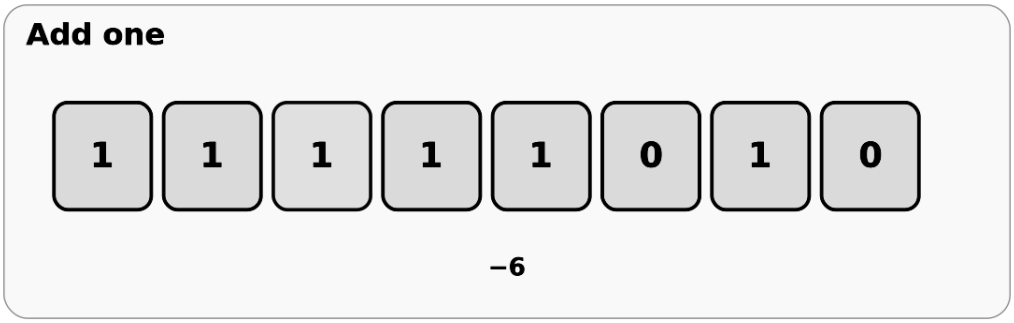
Problem

Model −6 in eight-bit two's complement.

Follow the worked method

1. Predict first — Write +6 as 00000110.
2. Check the method — Invert to 11111001.
3. Check the method — Add one to get 11111010, representing −6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Write 9 as 00001001.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

(a) Represent −19 as an 8-bit two's-complement number. [4] (b) Perform an arithmetic right shift by 1 of your result. State its signed decimal interpretation. [4]

Question 2

(a) Use De Morgan's law to simplify ¬(A ∧ ¬B). [3] (b) Explain how an eight-cell group of ones simplifies a three-variable Karnaugh map. [5]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A positive two's-complement fractional mantissa is 0.0011_2 with exponent 5. Normalise it, explaining each shift and why the value is unchanged.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An arithmetic right shift of −19 by one place produces −9.

Use key vocabulary from memory

1. Explain sign extension in your own words.

2. Explain De Morgan in your own words.

3. Explain normalisation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.